

MRATYUNJAY TRIPATHI

Senior Software Engineer | Payments & Lending Infrastructure | Gurugram, India
+91 88106 81526 • tripathi.mratyunjay.1@gmail.com • LinkedIn • GitHub

SUMMARY

Senior Software Engineer with 5+ years building payments and lending infrastructure for consumer credit at scale. Led the loan origination redesign that helped scale monthly disbursal from ₹30Cr to ₹100Cr, and integrated lending partners end to end across underwriting, disbursal, and UPI/NACH collections. Deep hands-on experience with Django microservices on AWS EKS, GitOps delivery via ArgoCD, and India Stack integrations (UPI mandates, NACH, Account Aggregator, DigiLocker, KYC). Promoted to Senior Software Engineer within two years; now the senior-most engineer on a 15-person tech team, leading a squad of four.

TECHNICAL SKILLS

Languages & Frameworks: Python, Django, Django REST Framework, JavaScript, React, SQL

Data & Messaging: PostgreSQL, Redis, RabbitMQ, Celery, multi-database architectures, query optimisation

Cloud & DevOps: AWS (EKS, EC2, S3), Kubernetes, Docker, ArgoCD, GitOps, GitHub Actions, Jenkins, CI/CD

Architecture: Microservices, REST API design, system design, distributed systems, event-driven processing

Domain: UPI, UPI Autopay mandates, NACH e-mandates, payment gateways, loan origination (LOS), underwriting, collections & settlement, KYC, Account Aggregator, DigiLocker

PROFESSIONAL EXPERIENCE

Kosh — Gurugram, India

Senior Software Engineer

January 2023 – Present

Full Stack Developer

February 2021 – December 2022

Lending & Payments

- Led the complete redesign of the Loan Origination System (LOS) serving 30K daily users, managing a team of 5 (3 backend, 2 frontend). The rebuild was a key enabler in scaling monthly disbursal from ₹30Cr to ₹100Cr.
- Integrated new lending partners end to end — underwriting, disbursal, collections via UPI and NACH, partner settlement, and loan closure — adding ₹4Cr per month of lending capacity.
- Built the Customer Data Platform (CDP) for KYC capture, validation, and retention, with a vendor-abstraction layer that allows providers (Signzy, Setu) for penny-drop account validation, Aadhaar masking, and Account Aggregator to be switched via a single configuration flag — removing vendor lock-in and shortening new-vendor onboarding.
- Automated loan refinancing for repeat borrowers, driving ₹50L+ in monthly refinanced disbursal with no manual intervention.
- Identified and remediated ₹20L+ of incorrect payment deductions across the collections pipeline, and hardened the reconciliation path to prevent recurrence.
- Integrated DigiLocker for PAN and Aadhaar retrieval during application, removing manual document collection from the underwriting flow and reducing turnaround time.
- Shipped GPS-based real-time location capture and the redesigned loan journey in the consumer app (10L+ downloads, growing ~50K/month).

Platform & Infrastructure

- Built the CI/CD pipeline on ArgoCD + GitHub for microservices running on AWS EKS across UAT and production — per-release image tagging with replica counts and resource limits continuously reconciled from Git (GitOps).
- Designed a hot-reload development environment on EKS giving each engineer a dedicated namespace; local code changes sync into the running pod in ~90ms and remote migrations sync back to local automatically, eliminating image rebuilds from the inner development loop.
- Migrated tables holding 10Cr+ (100M+) records off a primary-key dependency on the user model to independent primary keys, unblocking further schema evolution on the core lending data model.
- Architected a multi-database setup to partition data across databases, improving isolation and query performance as data volume grew.
- Built a complete loan underwriting console on Appsmith (JavaScript + PostgreSQL) in 48 hours to demo portfolio data to prospective lenders ahead of onboarding.

Infinocto Engineers Pvt. Ltd. — Noida Area, India

Software Developer (Internship)

January 2019 – February 2019

Ericsson India Global Services — Noida, India

Cloud Associate (Internship)

May 2019 – July 2019

EDUCATION

B.Tech, Computer Science — Jaypee Institute of Information Technology, Noida

2016 – 2020